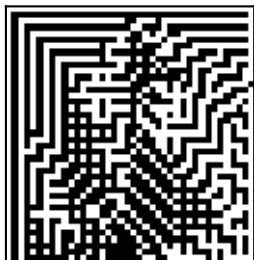


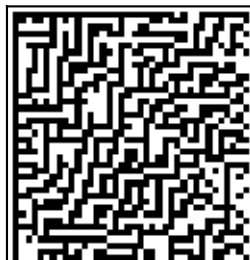
ES best-rule CA attractors (panels a-d) and failure modes (e-f)
Reproduced by deterministic Python mirroring GPU WGSL shader exactly (_ca_render.py ↔ gpu_engine_batched.js)

(a) manhattan-2 ckptScore=0.8233 [best]



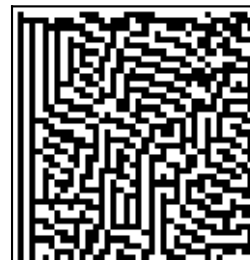
s=444 WxH=40x60
steps=300 iFs=True
raw_alive=1177/2400 (0.490)
canvas_alive=1177/2400 (0.490)
usedInverted=True proxy_orig=0.946 inv=0.974

(b) chebyshev-1 ckptScore=0.8095 [best]



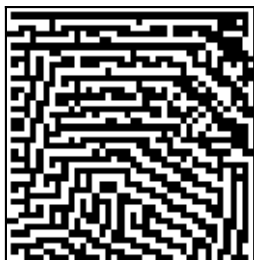
s=333 WxH=40x60
steps=300 iFs=True
raw_alive=1227/2400 (0.511)
canvas_alive=1227/2400 (0.511)
usedInverted=False proxy_orig=0.977 inv=0.976

(c) chebyshev-2 ckptScore=0.8080 [best]



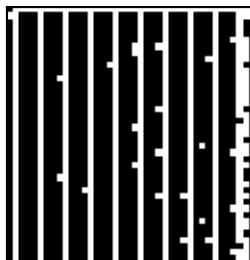
s=333 WxH=40x60
steps=300 iFs=True
raw_alive=1175/2400 (0.490)
canvas_alive=1175/2400 (0.490)
usedInverted=True proxy_orig=0.979 inv=0.979

(d) manhattan-4 ckptScore=0.7999 [best]



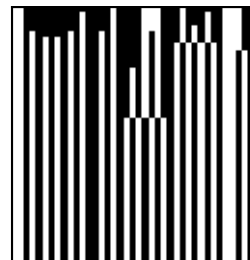
s=111 WxH=40x60
steps=300 iFs=True
raw_alive=1163/2400 (0.485)
canvas_alive=1163/2400 (0.485)
usedInverted=True proxy_orig=0.967 inv=0.968

(e) chebyshev-4 ckptScore=0.4244 [FAIL]



s=111 WxH=40x60
steps=300 iFs=True
raw_alive=745/2400 (0.310)
canvas_alive=745/2400 (0.310)
usedInverted=True proxy_orig=0.618 inv=0.620

(f) manhattan-1 ckptScore=0.4240 [FAIL]



s=444 WxH=40x60
steps=300 iFs=True
raw_alive=1203/2400 (0.501)
canvas_alive=1203/2400 (0.501)
usedInverted=False proxy_orig=0.998 inv=0.998